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Vertical Superior Vector Suspension of the Anterior Mobile Face with the V Plication Facelift: A Volumization and Lifting Technique

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Abstract
Background In most of the published plication techniques in face lift surgery, the vectors of plication are not entirely superiorly and vertically directed. The same applies with the deep plane, SMAS elevation techniques in the majority of which the vectors of traction are not superiorly vertically directed. The aging symptoms are mostly prominent at the anterior mobile face due to the gravity effect, and this is the area where attention should be focused to correct these symptoms following a face lift surgery.
Methods We have evaluated the results of 58 patients who underwent the V plication face lift between January 2021 to January 2024 in terms of demographics, procedure, surgical technique, adjuvant treatments, final outcomes and complications.
Results All 58 patients underwent primary face lift with the V plication for vertical superior traction and suspension of the facial tissues. The average age value was 64 years. Forty-six patients underwent concomitant upper and lower blepharoplasty, and 25 fat transfer of the upper periorbital region. There were one self-resolving neurapraxia of the right buccal branch of facial nerve and three minor self-absorbed localized blood collections.
Conclusion The extensive V plication face lift of the anterior face together with the progressively less extensive plication of the lateral face due to the shape of plication addressed satisfactorily the prominent symptoms of gravity effect on the face with the action of the vertical superior suspension of the facial tissues and corrected the malar deflation by repositioning the downwards sliding malar fat pad.
Level of Evidence V This journal requires that authors assign a level of evidence to each article. For a full description of these Evidence-Based Medicine ratings, please refer to the Table of Contents or the online Instructions to Authors www.springer.com/00266.

Keywords Facelift · V plication facelift · Anterior face plication · Malar fat pad · Midface volumization · Anterior mobile SMAS · Lateral face · Superior vector facelift

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Introduction

During the aging process, changes within both the superficial and deep fat compartments contribute to the ageing facial changes, noted by increased skin laxity, prominent folds, reposition and redistribution of fat and skin. The goal of facelift procedures is to modify and improve aging-related changes in the face and neck, rejuvenate the structure of the aging face and recreate a more youthful configuration.

Intrinsic factors, such as epidermal thinning, dermal elastosis, decreased elasticity and decreased collagen production, and extrinsic factors, such as photodamaged by chronic UVA exposure, smoking and radiation, contribute to aging of the different anatomical layers of face and neck [1–5]. In addition to changes in fat compartments and skin, the atrophy of the facial skeleton and the bony resorption lead to loss of facial volume and soft tissue support [6–10]. The goal of facelift procedures is to modify and improve aging-related changes in the face and neck, rejuvenate the structure of the aging face and recreate a more youthful configuration [2, 11].

Facelift is one of the most performed procedures in aesthetic plastic surgery [11]. Various techniques and approaches have been advocated through the years, and the literature is replete with published reports [2, 11]. These approaches include variances in skin incisions, varying degrees of skin undermining, skin lifts, SMAS (superficial musculoaponeurotic system), approaches (dual-plane approach, deep plane approach), etc. [12–22]. No operation or standard combination of operations meets the needs of every patient [11, 14]. Moreover, in the majority of the SMAS plication techniques that have been published the vectors of plication are not entirely superior vertical directed. The same applies with the deep plane, SMAS elevation techniques. In most of them the vectors of traction are not superior vertical.

We present our evolved approach which is focused to the anterior mobile face aiming to correct the symptoms of the centripetal force of gravity along with its effects on the face, by applying the opposite centrifugal force to support the affected tissues. This technique incorporates treatments of all the facial structures of facial aging, being appropriate even for severe cases, such as heavier faces, with long lasting results. The V-shaped SMAS (superficial musculoaponeurotic system) plication facelift effectively addresses the mobile part of the anterior face, where mainly aging and gravity symptoms are evident, and progressively ends up with a less extended plication of the lateral face, where the SMAS is less mobile but also the aging signs are not so severe and prominent. Additionally, the V-SMAS plication with the double V technique contributes to repositioning of the sliding malar fat pad vertically superiorly, in a higher position, as well as improvement of the jaw line contour. The V-shaped SMAS plication facelift can be customized in terms of angle (the aperture of V limbs depending on the degree of laxity), vectors and width, according to patients' indications to accomplish reshape and restoration of the lax tissue of the anterior face and of course versatility in addressing the different ageing manifestations. Therefore, this technique that will be described in more details below is safe, reproducible, efficient and less complicated for the facial tissues with exceptional surgical outcomes with respect to SMAS elevation.

Objective: To focus on correcting the main symptoms of laxity due to gravity effect on the anterior mobile face with a vertical, opposite, superior direction force which can neutralize the effect of the centripetal gravity force which is the main reason of the aging signs of the face

Methods

Anatomy Pearls

The mobile anterior face is separated from the relatively fixed lateral face. The lateral face is fixed in place by the platysma auricular fascia (PAF) and is separated from the anterior one by a vertical line corresponding to retaining ligaments: temporal adhesion (orbital ligament), lateral orbital thickening (superficial canthal tendon), zygomatic ligaments, masseteric ligaments, and mandibular ligament [2] . The anterior face functionally is adapted for facial expressions, and the lateral face overlies the masticatory structures. After the early studies of Skoog in 1974 and later Peyronie talking about musculoaponeurotic system, the term SMAS has been extensively used for every face lift technique described [16, 23]. Recently, there has been debate about the existence of such a structure as a true aponeurosis in the central face. Cadaveric studies from Mendelson's laboratory demonstrate that the only attachment between the skin and the deep fat within the deep fascia is the *retinacula cutis superficialis*; the findings correspond to Skoogs' original clinical observations that there is a transition from the deep fascia of the lateral face into the subcutaneous tissue of the mobile anterior face. It is now important to understand that the thick flap of soft tissue anterior to the masseter lacks a true aponeurosis and that the strength of the flap to maintain fixation is achieved with the retinacula cutis superficialis [24]. The platysma auricular fascia as described by Mendelson is a very rigid element that helps support and leaving it intact can be used for suture fixation [2]. This is a difficult dissection because it is not a natural plane of separation. Moreover, since the objective of face lift aesthetic surgery is to correct laxity in the mobile anterior face, the dissection in the lateral face is of secondary importance [2]. The ageing symptoms and therefore laxity are more prominent in the anterior face and mainly due to the vertical centripetal force of gravity (malar mount sliding, marionette lines, jowls, etc.) (Figs. 1, 2) However, the vectors of most of the SMAS elevation (deep plane) techniques, except the high SMAS, are not vertical and opposite to the gravity force. Even in the case of high SMAS nevertheless, the distance of traction is quite long (zygoma to jowls), and therefore, according to the Hook's law and Lawrence Evans principle the lifting effect is minor at the distant point of the traction force.

Moreover, some of the deep plane techniques (like composite flaps) do not give the versatility of moving the

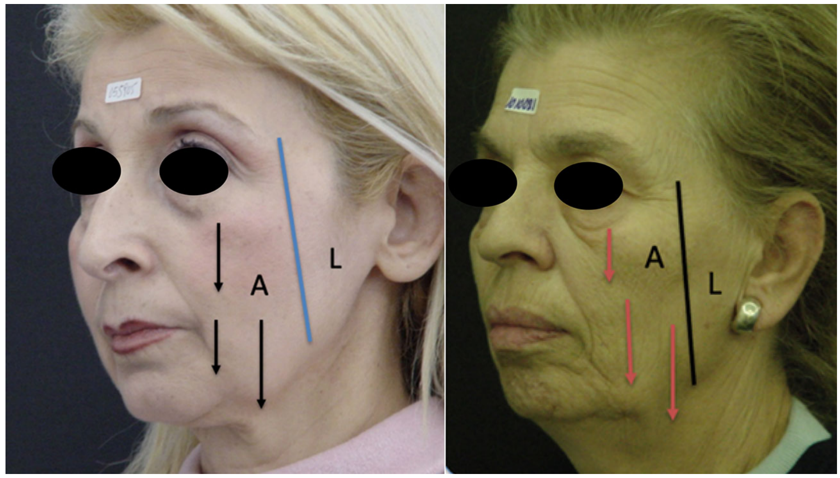

Fig. 1 Location of gravity symptoms of aging in the anterior face (marionette lines, jowls, sliding of the malar fat pad, deflation of upper midface). A: anterior face, L: lateral face

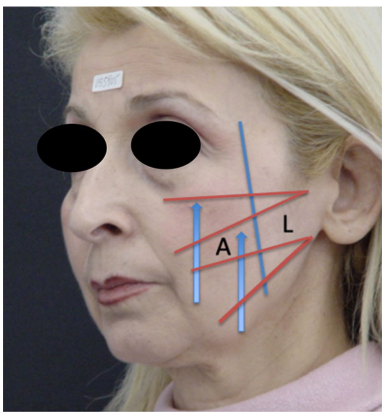

Fig. 2 Schematic drawing of the V patterns to address the anterior face

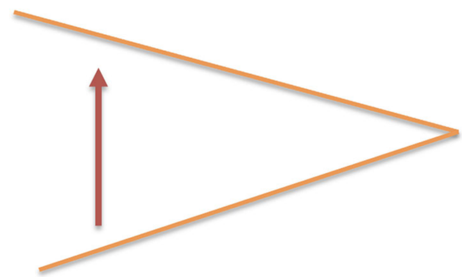

Fig. 3 Application of centrifugal force of the V pattern on the facial tissue at the anterior face (longer limb distance) and lateral face (shorter limb distance)

SMAS and skin flap to different directions for optimization of final outcomes, since skin and SMAS flaps must move en bloc and towards one direction only.

To correct the ageing symptoms of gravity, we must apply the vertical opposite to gravity force at the anterior face (vertical superior centrifugal vector).

Preoperative Markings and Surgical Technique

Over the last 5 years, we performed a subcutaneous flap elevation facelift with SMAS plication according to preoperative markings which comprise two V-shaped plication areas. These are totally customized as far as the angles, the position and the desirable vectors are concerned, always according to patients' needs (Figs. 2, 3). The objectives are to plicate more and superiorly vertically, in the anterior face than the lateral, due to different mobility of the SMAS and to reduce the distance of traction to achieve better and long-standing effect. The desirable youthful contour is accomplished by the elevation of the malar fat pad and reposition of the superficial fat in the mobile medial midface. It must be pointed out here that this technique does not only offer a SMAS plication of the soft tissue but also stable and strong suspension of the plicated tissues on the periosteum of the zygoma.

The superior V is customized according to patient's need to elevate the sliding malar fat pad superiorly, while the aperture of both V angles depends on the degree of skin laxity and malar fat pad sliding. The superior V is marked on the upper face where its superior limb sits along the lower aspect of the zygomatic arch and the inferior limb in customized angulation including the body of the malar fat pad (Figs. 4, 5, video 1). The aim is to relocate the malar fat pad in higher position and offer projection to the zygomatic region which has been deflated due to the downward sliding of the malar fat pad from gravity effect and fat atrophy.

This relocation is achieved by anchoring the plicated tissue on the periosteum of the zygomatic arch and not just plicating tissue to tissue as in most of the plication techniques described.

The inferior V is marked on the lower face where its inferior limb sits on the mandibular line, and the superior one extends from 1–2 cm from oral commissure to

Fig. 4 Marking of the superior and inferior V on the skin and on the SMAS after flap dissection

Fig. 5 Detailed marking of the zygomatic arch, malar fat pad, superior and inferior V, vertical superior vector direction of plication. See also video 1 for details.

preauricular region (Figs. 4, 5, video 1). As the plication of the tissue starts from the superior V, the extend of limb distance of the inferior V and amount of plication depend on the remaining mobility and laxity of the lower facial tissue, since the superior V plication also addresses the laxity of the inferior part of the face to a certain degree.

The preferred incision starts from the junction of the ascending crus of the helix and the cheek and continues inferiorly placed in the pretragal sulcus and then around the earlobe and subsequently courses superiorly where it is placed slightly up on the ear; the scar is prone to migration and is best hidden finally in the depth of the sulcus. The incision travels posteriorly along the occipital hairline (Fig. 6). The dissection of the preauricular and anterior face starts with scalpel for the first 1–2 cm and then switched to scissors to a complete elevation of a subcutaneous flap (Fig. 7). The preauricular flap undermining extends according to the degree of the aging changes. Nevertheless, it is important to release the subcutaneous attachment (not sub-SMAS) of the mandibular retaining ligament to achieve better and longer-lasting contouring of the mandibular line. Undermining is always performed under direct vision with the use of fiber optic retractor to assure the integrity of facial nerve branches, performing dissection over the SMAS. The skin flap should always be of an appropriate thickness to ensure its viability. In the post-

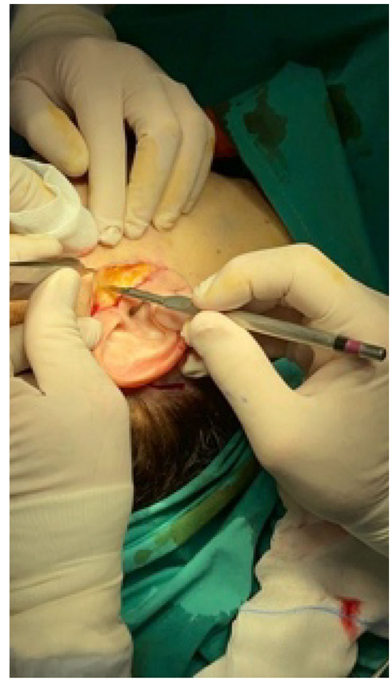

Fig. 6 Initial dissection of subcutaneous flap with scalpel

auricular area, the flap is firmly attached to the deep cervical fascia of the sternocleidomastoid and the mastoid. This is the most common location for skin flap necrosis, so the flap is raised under direct vision, keeping the dissection, against the underlying deep fascia to maintain flap thickness. As the dissection continues in the subcutaneous plane, the great auricular nerve is protected.

After dissection is completed, marking of the V-SMAS plication areas is performed with a sterile skin marker according to the patient's indications. The two Vs are drawn customizing the angles, the extent of their limbs and the exact orientation to accomplish the desired outcome by aiming in applying vertical centrifugal force in the anterior mobile face region. The plication of the limbs of each V is made with Monocryl 3/0 absorbable suture. The plication of the upper V starts first with a Monocryl 3/0 absorbable suture in figure eight mode from malar fat to the malar periosteum for additional support of the new position of the malar fat pad. Suturing continues with the same suture in over-and-over mode back and forth to stabilize the

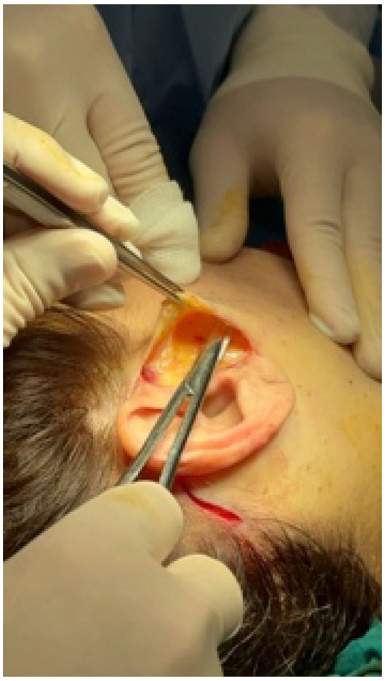

Fig. 7 Further dissection and elevation of the flap with scissors

plication. The lower V is then plicated with same sutures in an over-and-over locked mode. Both plications should end up in straight lines after bringing together the limbs of the Vs (Fig. 8) (video 2, 3). Attention should be taken when plicating the malar fat pad to avoid injury of the zygomatic branch of the facial nerve. Needle bites of the malar fat pad should include the fat pad itself with the overlying retinacula cutis superficialis (or otherwise mentioned traditionally as SMAS) and anchored to the malar periosteum in a superficial direction to avoid injury of the zygomatic branch. This maneuver has been proved adequate to support the new location of the malar fat pad and is stabilized with further suturing in a superficial over-and-over mode of the remaining gap between the V limbs (video 2, 3).

It is of utmost importance to avoid excessive tightening of the superior V limbs during plication. This is achieved by prudently marking the superior V shape, so that the malar fat pad will be relocated to the correct position and with no excessive tension. This maneuver will protect from overcorrecting of the malar fat pad which can lead to prolonged recovery and patient's discomfort.

The plication of the inferior V is safer and performed with over-and-over or figure eight mode sutures to also end up in a straight line (Fig. 8 video 4).

The undermined skin flap is redraped in a lateral direction, and the skin flap is fixed at the junction of the ascending helical crus with the cheek and at the apex of the retro auricular incision. After the placement of these two tension-bearing sutures, trimming of the overlapping flap is done and an incision of the skin flap is made in a way that the proper inset of the ear lobe will leave no possibility for the scar to be visible. It is important that the earlobe is detached up to the cartilaginous joint and properly inset to prevent pixie—ear deformity and tension. Finally, a hollow silicone drainage tube is placed at the lateral end of the postauricular incision to drain in the retroauricular dressing. Both drains are removed after 72 hours (Fig. 9).

Skin closure is accomplished with fine absorbable 4/0 Monocryl suture and non-absorbable 6/0 and 5/0 prolene suture, with subcuticular and over-and-over suturing, respectively (Fig. 9). Postoperative care includes the necessary bandaging and light compression garments with follow-up of the patient 3-5-7- and 10-days post-op.

In certain cases as shown in Table 1, the neck was also addressed as a first step of the operation according to the indication of each patient with platysma band plication, subplatysmal fat excision or supraplatysma fat liposuction, anterior digastric corset/shaving or combination of them.

Finally, fat transfer in accordance with the basic principles of fat grafting by harvesting fat from the abdominal area was performed only for the upper eyelid fullness restoration in 25 out of the 46 patients who underwent blepharoplasty together with the V plication facelift technique.

Results

The V facelift was performed on 58 patients, including 51 females and 7 males (Table 1). The average patient age when facelift was performed was 64 years old with a range of 53–72. All patients underwent a primary facelift. The V facelift was performed in combination with blepharoplasty in 79% of the patients and fat transfer in 43% of them. Patients were followed for an average of 22 months postoperatively with a range of 12–31 months. Examples of patients are shown in Figs. 10, 11, 12, 13, 14.

The V face lift was proved to be a safe, reproducible and simple technique, in correcting the main symptoms of gravity in the anterior face. No major complications were noted apart from three cases of unilateral self-absorbed mild blood collection with no necessity of surgical drainage. One patient presented with a neurapraxia of the buccal branch of the right part of the face, most probably due to injury of the branch by the suture used to support the malar fat pad in higher position at the zygomatic region. The patient presented with weakness of the upper lip which gradually self-resolved in a period of 3 months (Table 2).

Patients' satisfaction was high, mainly mentioning the fast recovery postoperatively and return to everyday activities in due course.

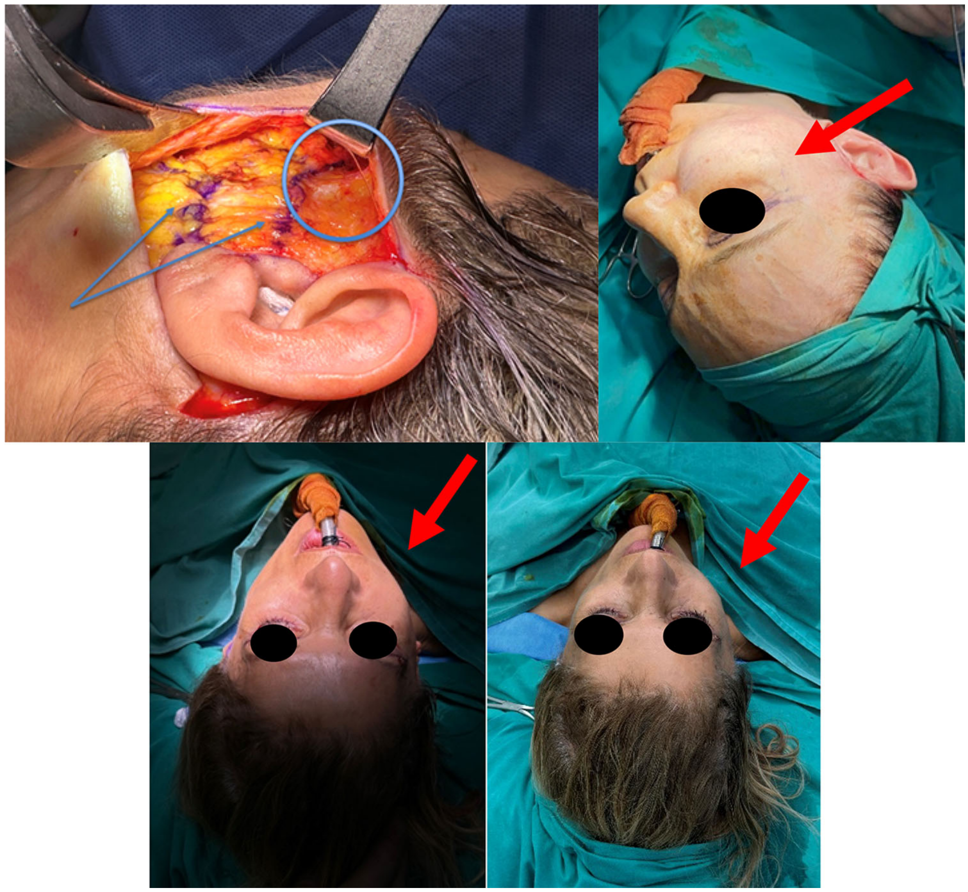

Fig. 8 Intraop photo of conversion of the V pattern to straight line after plication of the limbs (blue arrows) and relocation of the malar fat pad (red arrows)

Fig. 9 Placement of incision in the pretragal sulcus and closure of skin and drain in place

Discussion

The primary author has been applying different techniques in face lift surgery for almost 30 years including high SMAS with extended preparotid elevation and resection, anterior SMAS elevation, composite skin-SMAS flap elevation, direct SMASectomy and SMAS plication in different vectors. The follow-up of patients in the long term has not shown dramatic differences in the final long-standing results with more complicated deep plane techniques compared to the more simplified ones like SMAS plication alone. This has been reported also by several authors in the past [25, 26].

Patients' mentality and expectations are also another factor which is related to ethnicities, demands for natural results postoperatively, minimal correction of aging signs and temperament. With the invasion of technology in aesthetic surgery and popularity of noninvasive treatments for the facial rejuvenation, in our practice, we have noticed in the last 10 years, an increased demand for simpler operations, and even more, shift of patients in request of non-surgical interventions. Moreover, younger age individuals seek correction of aging signs, therefore presenting to the practice without severe aging manifestations but at the same time requesting reverse of the aging signs even at earlier stages of laxity and gravity effect on the face.

In addition to the above observations, taking into account the longer postoperative recovery, the rate of minor or major complications and the final outcome of

Table 1 Patients' data

Total patients	58
Average age	64 yrs
Age range	52-71 yrs
Females	52
Males	6
Primary operation	58
Blepharoplasty	46(79%)
Fat transfer	25(43%)
Neck procedure (platysma plication, anterior digastric corset, neck liposuction or fat excision)	37(63%)

Fig. 10 64 years old patient operated with the V-SMAS plication facelift, upper and lower blepharoplasty, two and a half years post-op. Note the volumization of the zygomatic region, the correction of the jowle area and the correction of aging signs of the anterior mobile face. Patient underwent also a rhinoplasty 6 months after the face lift operation

Fig. 11 56 years old patient operated with the V-SMAS plication facelift, two years post-op. Correction of the aging signs of the anterior face are prominent

more complicated interventions for the rejuvenation of the face, we have gradually shifted into simpler but equally efficient approaches in facelift surgery by introducing the V-shape plication facelift, which is targeting in correcting the main complaints of patients focused in the anterior face, with significantly less downtime, higher safety of patients and reproducible, high standard results.

The superior vector plication in face lift seems to receive wide acceptance in the recent years [27], and our V technique is introduced to specify in terms of physics and tissue dynamics the importance and efficiency of the V

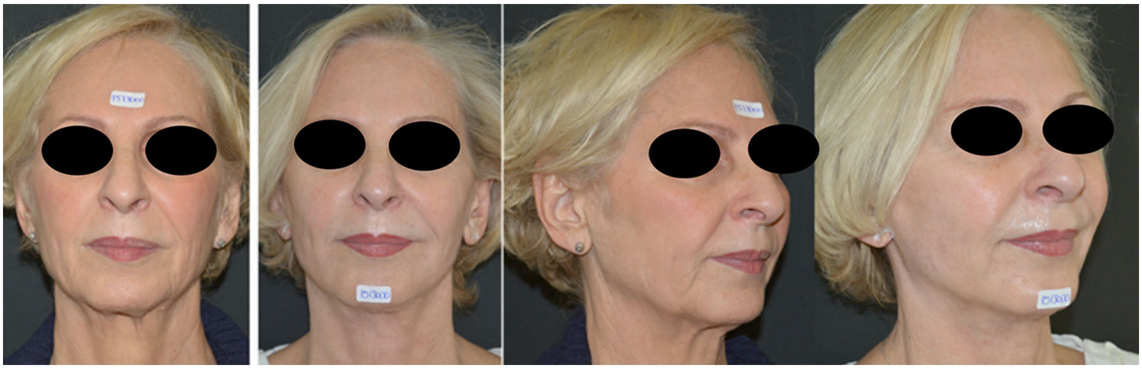

Fig. 12 3 years post-op result of a patient operated with the V-SMAS plication facelift. Stable result of correction of aging signs of the anterior face and volumization of upper face

Fig. 13 Two and a half years post-op result of a 54 years old patient with mild pre-op laxity of the anterior face. Notice the sufficient volumization of the mid face post-op

Fig. 14 65 years old patient with excessive aging signs of the anterior face and neck. The V plication face lift was applied together with platysma bands correction. Result after one and a half year

pattern for plication. The understanding of the dynamic effect of the centripetal force of gravity and accurate opposite direction of centrifugal force for the correction of the gravity effect symptoms of the anterior face is adding significant importance to this technique.

Further follow-up and data collection of patients will continue in order to document the stability of the midface volumization and the lifting effect of our technique despite the promising results achieved so far.

Table 2 Complications

Motor neurapraxia buccal branch	1
Minimal self-absorbed blood collection	3
Postauricular minor skin sloughing	2
Post operative irregularities	4
Suture granulomas	3
Suture protrusion at preauricular region	4

Conclusion

Facial rejuvenation in patients with ageing symptoms requires restoration of lost volumes and jaw line contouring. According to our experience, SMAS plication with the double V technique even in more severe cases and heavier faces has worked the same well as elevation of the SMAS and other sub-SMAS techniques. The SMAS flap elevation and fixation may not provide longer-lasting results over plication in certain cases. On the other hand, the V-SMAS plication and periosteal suspension improves malar fat pad re-insertion in higher position and the jaw line by correcting the jowls along it. In addition to the above, our experience showed that the subcutaneous attachment of the mandibular retaining ligament must be released to achieve better and longer-lasting contouring of the mandibular line when doing V-SMAS plication. Finally, the V pattern can be customized in terms of angle, vector and width depending on patient indications to achieve repositioning of the lax tissue of the anterior face and offers versatility in addressing the different ageing manifestations. The clinical observation supports the use of our technique to all age groups with minimal complications and short recovery time.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00266-025-04661-x.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest to disclose.

Human Participants This article does not contain any studies with human participants or animals performed by any of the authors.

Informed Consent For this type of study, informed consent is not required.

References

1. Schenck TL, et al. The functional anatomy of the superficial fat compartments of the face: a detailed imaging study. Plast Reconstr Surg. 2018;141(6):1351–9.
2. Warren RJ, Aston SJ, Mendelson BC. Face lift. Plast Reconstr Surg. 2011;128(6):747e–64e.
3. Cotofana S, Lachman N. Anatomy of the facial fat compartments and their relevance in aesthetic surgery. J Dtsch Dermatol Ges. 2019;17(4):399–413.
4. Donofrio LM. Fat distribution: a morphologic study of the aging face. Dermatol Surg. 2000;26(12):1107–12.
5. Swift A, et al. The facial aging process from the “inside out.” Aesthet Surg J. 2021;41(10):1107–19.
6. Aston SJ, Steinbrech DS, Walden JL. Aesthetic plastic surgery e-book. Saunders; 2012.
7. Bartlett SP, Grossman R, Whitaker LA. Age-related changes of the craniofacial skeleton: an anthropometric and histologic analysis. Plast Reconstr Surg. 1992;90(4):592–600.
8. Pessa JE. An algorithm of facial aging: verification of Lambros's theory by three-dimensional stereolithography, with reference to the pathogenesis of midfacial aging, scleral show, and the lateral suborbital trough deformity. Plast Reconstr Surg. 2000;106(2):479–88.
9. Pessa JE, Chen Y. Curve analysis of the aging orbital aperture. Plast Reconstr Surg. 2002;109(2):751–5.
10. Shaw RB Jr, et al. Aging of the mandible and its aesthetic implications. Plast Reconstr Surg. 2010;125(1):332–42.
11. Alhithlool AW, et al. Top 50 highly cited publications in facelift surgery: a 50-year bibliometric analysis review. Aesthetic Plast Surg. 2024;48(3):312–23.
12. Baker DC. Lateral SMASectomy. Plast Reconstr Surg. 1997;100(2):509–13.
13. Baker DC. Minimal incision rhytidectomy (short scar face lift) with lateral SMASectomy: evolution and application. Aesthet Surg J. 2001;21(1):14–26.
14. Farhadieh R, et al. Plastic surgery—principles and practice, vol. 1. Amsterdam: Elsevier Health Sciences; 2021.
15. Marten TJ. High SMAS facelift: combined single flap lifting of the jawline, cheek, and midface. Clin Plast Surg. 2008;35(4):569–603.
16. Mitz V, Peyronie M. The superficial musculo-aponeurotic system (SMAS) in the parotid and cheek area. Plast Reconstr Surg. 1976;58(1):80–8.
17. Mustoe TA, Park E. Evidence-based medicine: face lift. Plast Reconstr Surg. 2014;133(5):1206–13.
18. Rohrich RJ, et al. The individualized component face lift: developing a systematic approach to facial rejuvenation. Plast Reconstr Surg. 2009;123(3):1050–63.
19. Stuzin JM, et al. Cervical contouring in face lift. Aesthet Surg J. 2002;22(6):541–8.
20. Tonnard P, Verpaele A. The MACS-lift short scar rhytidectomy. Aesthet Surg J. 2007;27(2):188–98.
21. Trussler AP, et al. The frontal branch of the facial nerve across the zygomatic arch: anatomical relevance of the high-SMAS technique. Plast Reconstr Surg. 2010;125(4):1221–9.
22. Wan D, Small KH, Barton FE. Face Lift. Plast Reconstr Surg. 2015;136(5):676e–89e.
23. Skoog T. Plastic surgery: new methods and refinements. Saunders; 1974.
24. Minelli L, Mendelson B, et al. The superficial musculoaponeurotic system: does it really exist as an anatomical entity?
25. Chang S, Pusic A, Rohrich R. A systematic review of comparison of efficacy and complication rates among face-lift techniques. Plast Reconstr Surg. 2011;127(1):423–33.
26. Becker F, Bassichis B. Deep plane facelift vs superficial SMAS plication face lift. Comparative study Arch Facial Plastic Surgery. 2004;6:8–13.
27. Grotting J. Superior vector plication of the mobile anterior midface: the delta facelift. Aesthetic Surg J. 2024;44(2):144–59.